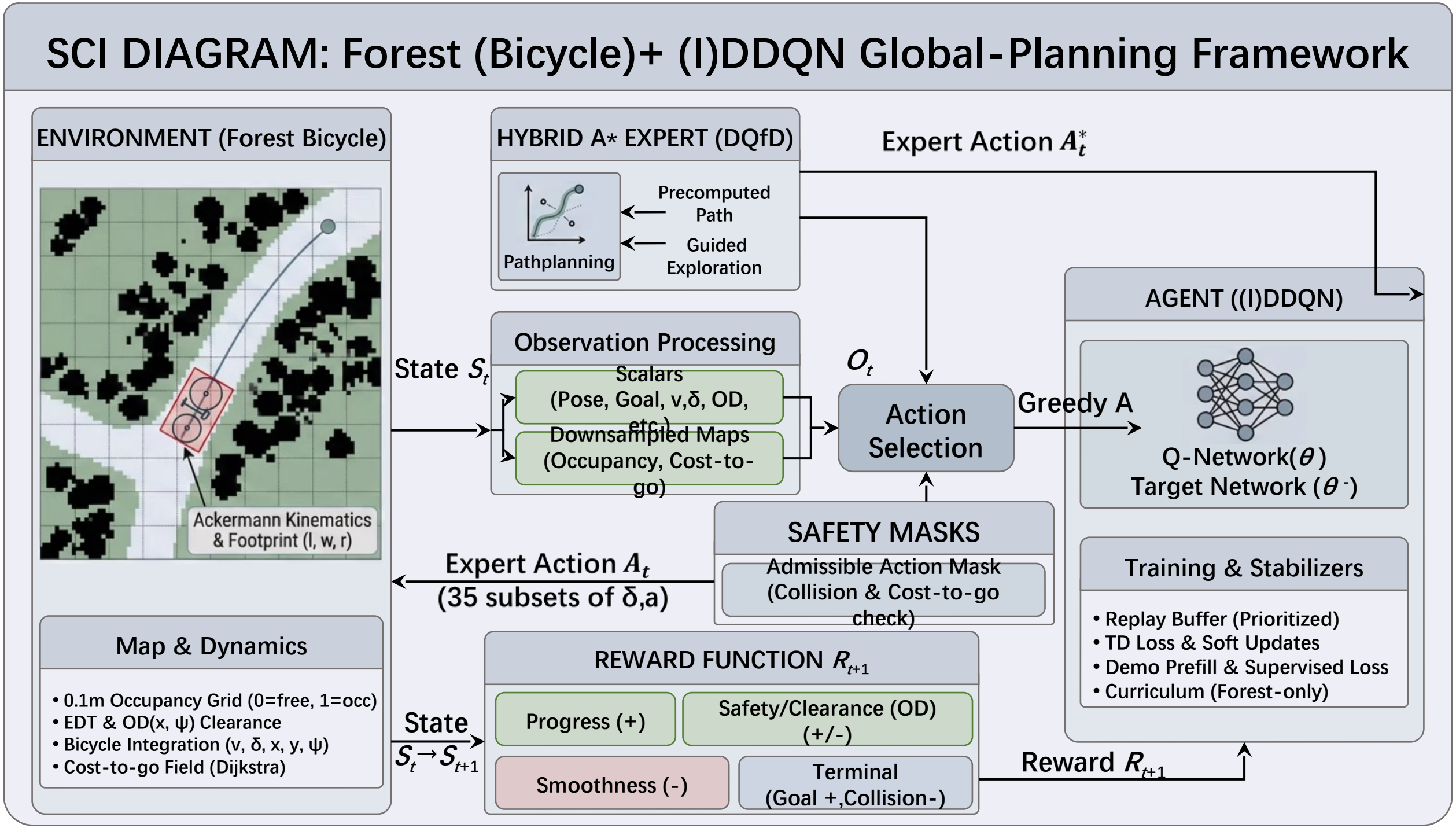
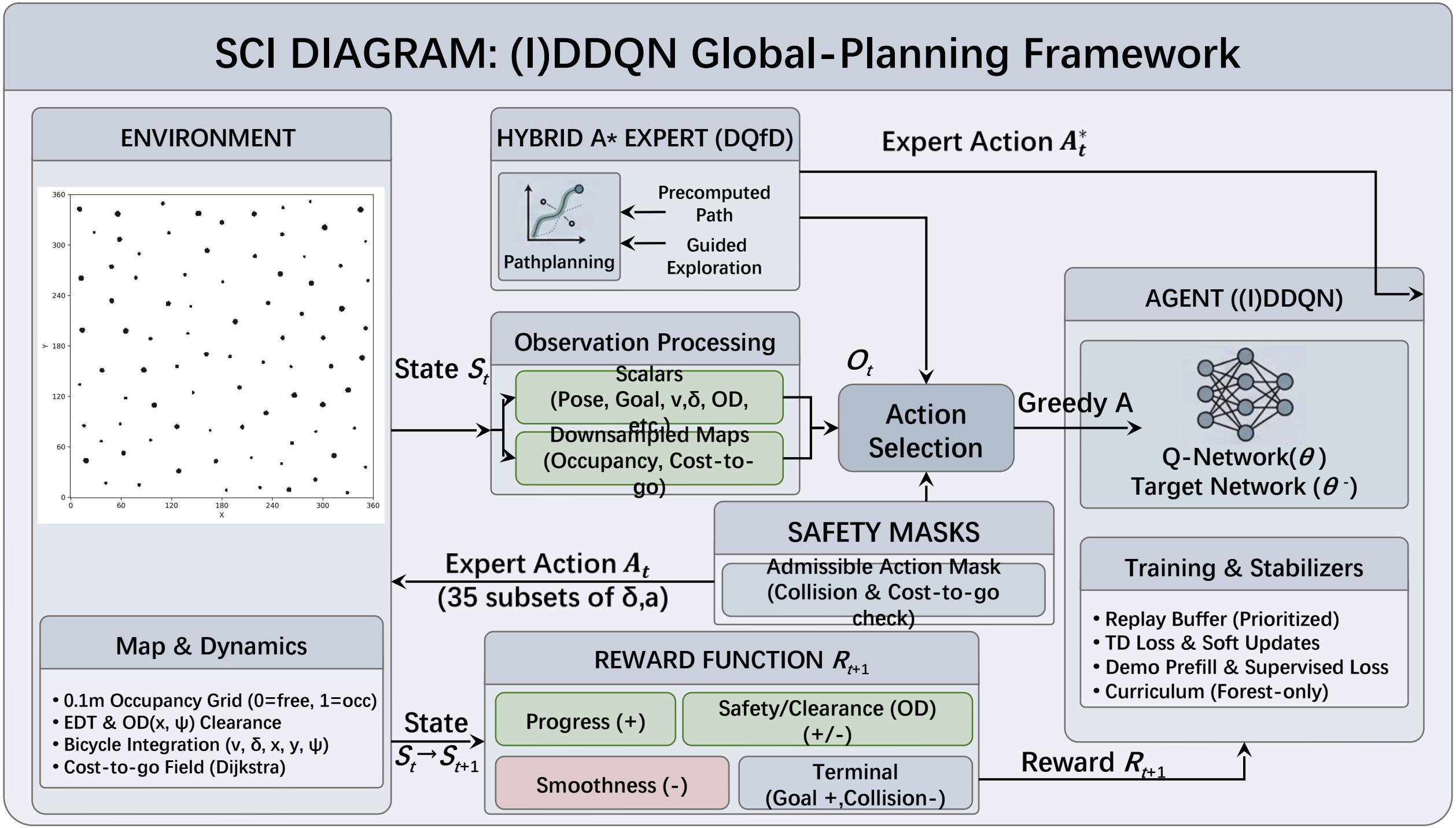
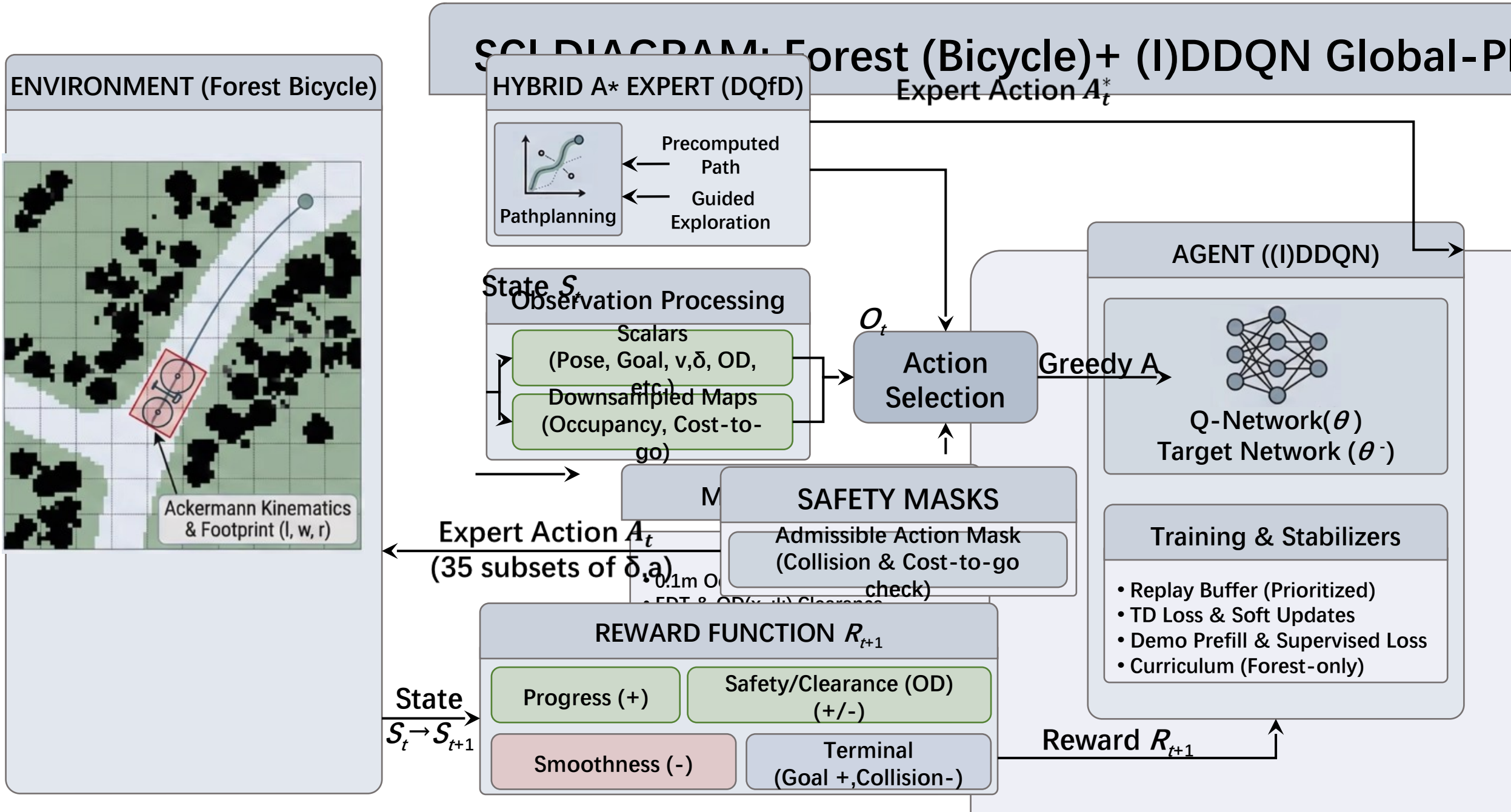
SCI DIAGRAM: Forest (Bicycle)+ (I)DDQN Global-Planning Framework

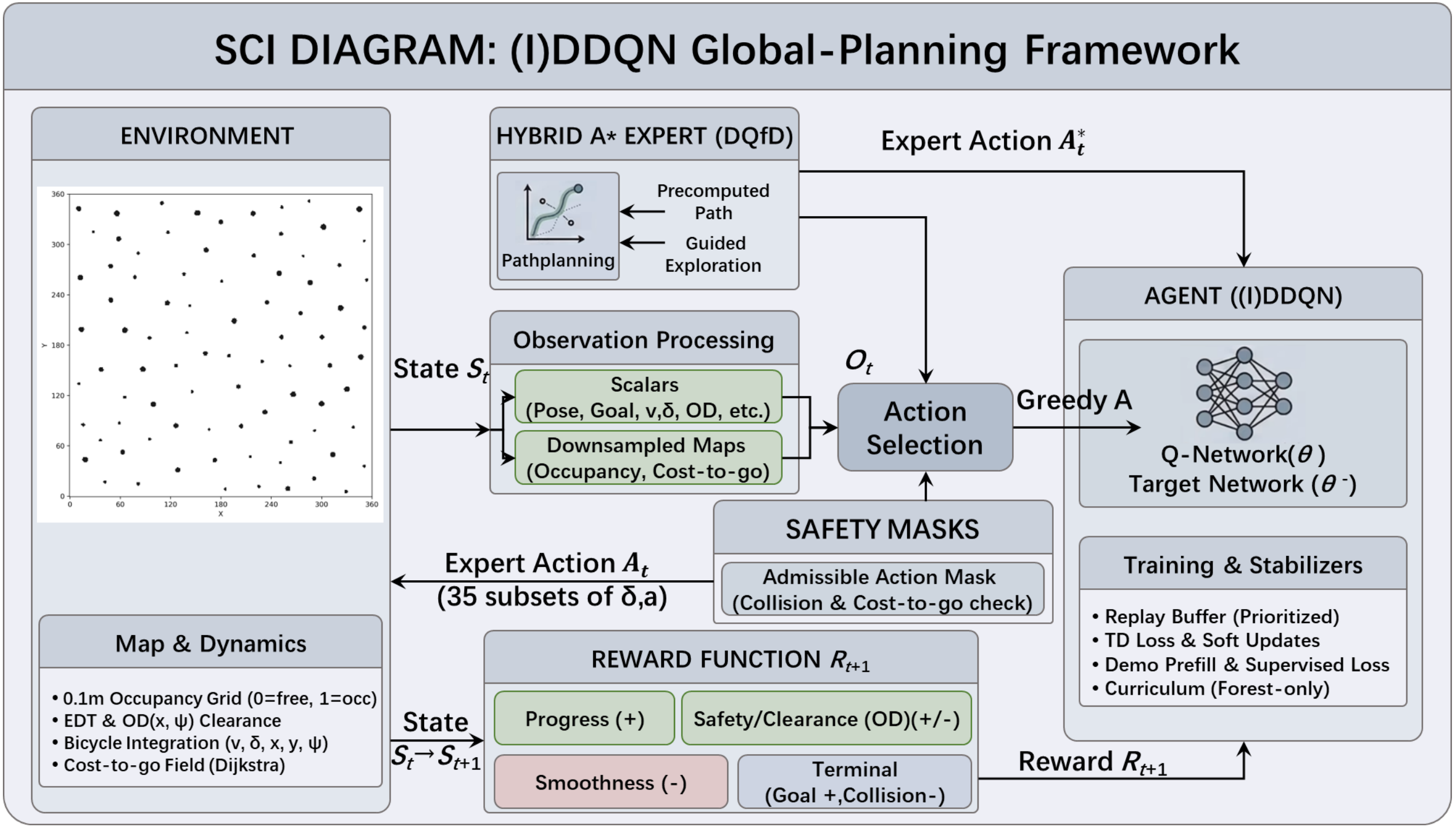


SCI DIAGRAM: (I)DDQN Global-Planning Framework




SCI DIAGRAM: Forest (Bicycle)+ (I)DDQN Global-Pl
ENVIRONMENT (Forest Bicycle)
Ackermann Kinematics & Footprint (l, w, r)
HYBRID A* EXPERT (DQfD)
Pathplanning
Precomputed Path
Guided Exploration
Expert Action A_t^*
State S_t
Observation Processing
Scalars (Pose, Goal, v,δ, OD, etc.)
Downsampled Maps (Occupancy, Cost-to-go)
O_t
Action Selection
Greedy A
AGENT ((I)DDQN)
Q-Network(θ)
Target Network (θ^-)
Training & Stabilizers
• Replay Buffer (Prioritized)
• TD Loss & Soft Updates
• Demo Prefill & Supervised Loss
• Curriculum (Forest-only)
SAFETY MASKS
Admissible Action Mask (Collision & Cost-to-go check)
Expert Action A_t
(35 subsets of δ,a)
REWARD FUNCTION R_{t+1}
Progress (+)
Safety/Clearance (OD) (+/-)
Smoothness (-)
Terminal (Goal +,Collision-)
State
$S_t \rightarrow S_{t+1}$
Reward R_{t+1}

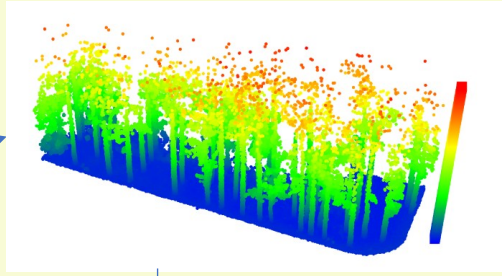
SCI DIAGRAM: (I)DDQN Global-Planning Framework



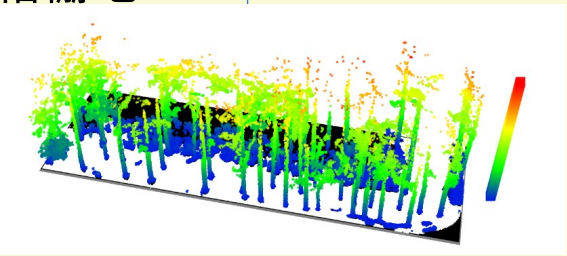
地面无人车平台

森林场景格栅地图的建立

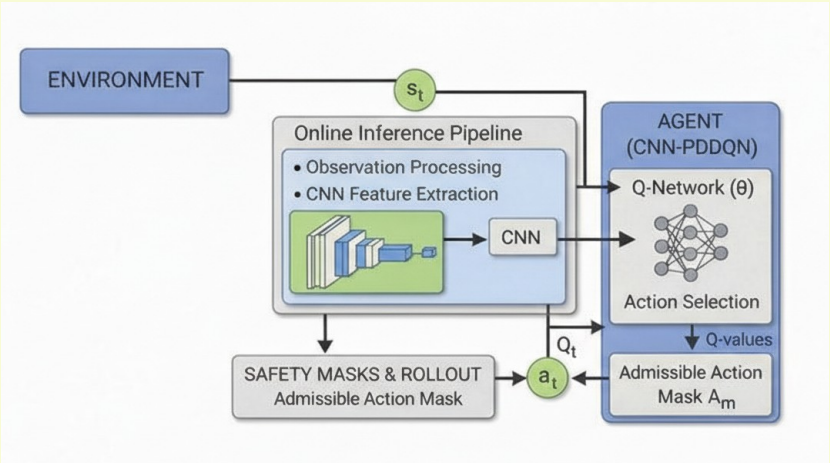
Slam 得到点云图



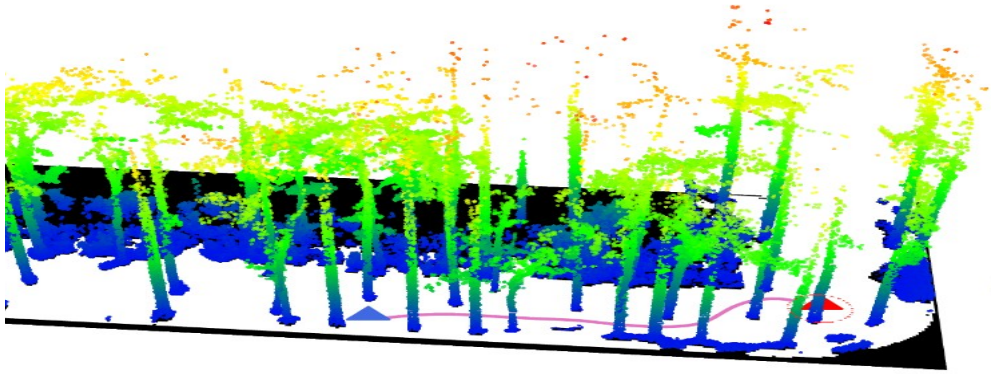
转化为格栅地图



CNN-PDDQN 运动规划算法设计



真实场景路径规划



知识目标

夯实专业基础

问题一

学生进行知识迁移和创新应用的能力不足